Dynamics of Self-propelled particles in laminar flow

Under the Guidance of:

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Overview and introduction

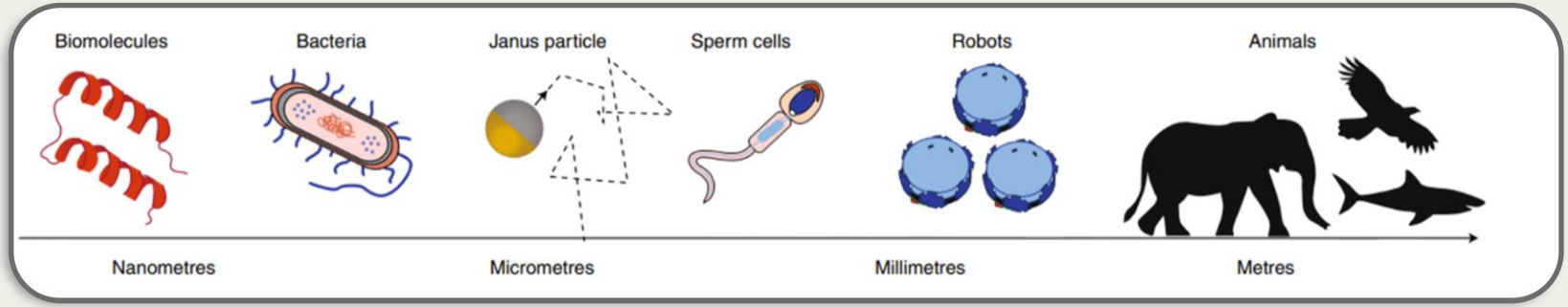


Image Credit : Cichos, F., Gustavsson, K., Mehlig, B. *et al.* Machine learning for active matter.

Active matter systems across scales: From nanoscale biomolecules to microscale particles like bacteria and Janus particles, to millimetre-scale robots, and up to meter-scale animals and organisms, demonstrating the universality of active matter dynamics.

What is Active matter?

- Systems composed of particles capable of self-propelled motion.
- Consume energy to produce autonomous movement.

Why Active Matter is Important ?

- Unlike passive particles, active particles generate their own motion, leading to unique and complex collective behaviors.
- Interaction with external environments (fluid flows, fields) creates distinct spatial and temporal patterns.

Cellular Flow Field used in this study

Dynamics of the Self-Propelled Particles

$$\dot{\vec{r}} = \vec{v}_f(\vec{r}, t) + v_p \hat{p}$$

$$\dot{\hat{p}} = \frac{1}{2}(\vec{\omega} \times \hat{p}) + \alpha(\hat{p} \cdot \mathbf{E})(\mathbf{I} - \hat{p}\hat{p})$$

$\vec{v}_f$: Background fluid velocity

v_p : Self-propulsion speed

$\hat{p}$: Orientation unit vector

$\vec{\omega}$: Vorticity

$\mathbf{E}$: Rate-of-strain tensor

α : Shape parameter

In 2D

$$\dot{x} = v_x(x, y, t) + v_s \cos\theta$$

$$\dot{y} = v_y(x, y, t) + v_s \sin\theta$$

$$\dot{\theta} = -\alpha \cos x \cos y \sin(2\theta) + \sin x \sin y$$

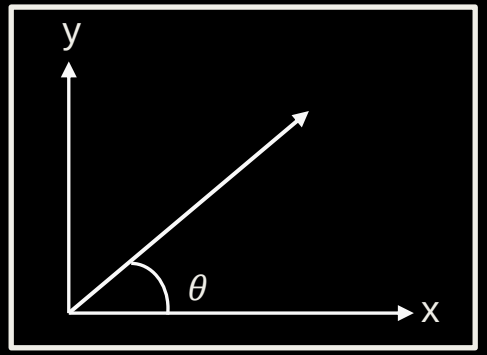
x, y : Position of the particle in 2D space

$v_x(x, y, t)$: x-component of the fluid velocity field at the particle's position

$v_y(x, y, t)$: y-component of the fluid velocity field at the particle's position

v_p : Self-propulsion speed(constant magnitude of active motion)

θ : Orientation angle of the particle relative to the x-axis

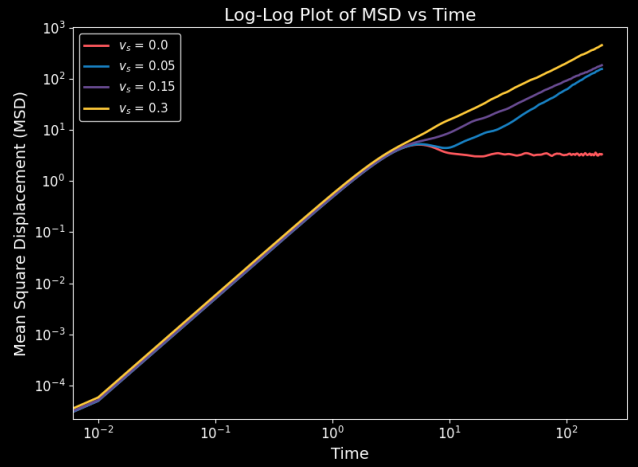


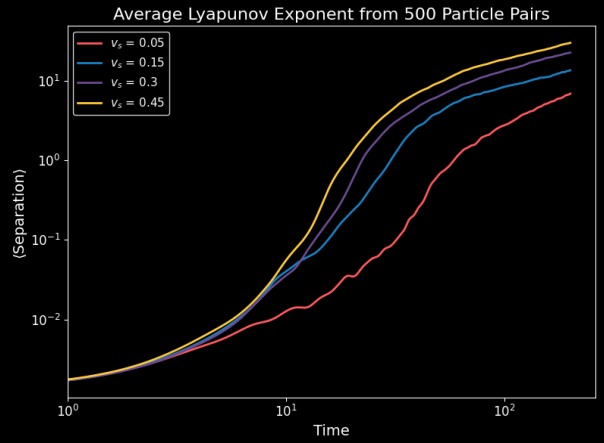
Recap of Mini Project

- Simulated the motion of spherical self-propelled particles in a 2D cellular flow field.

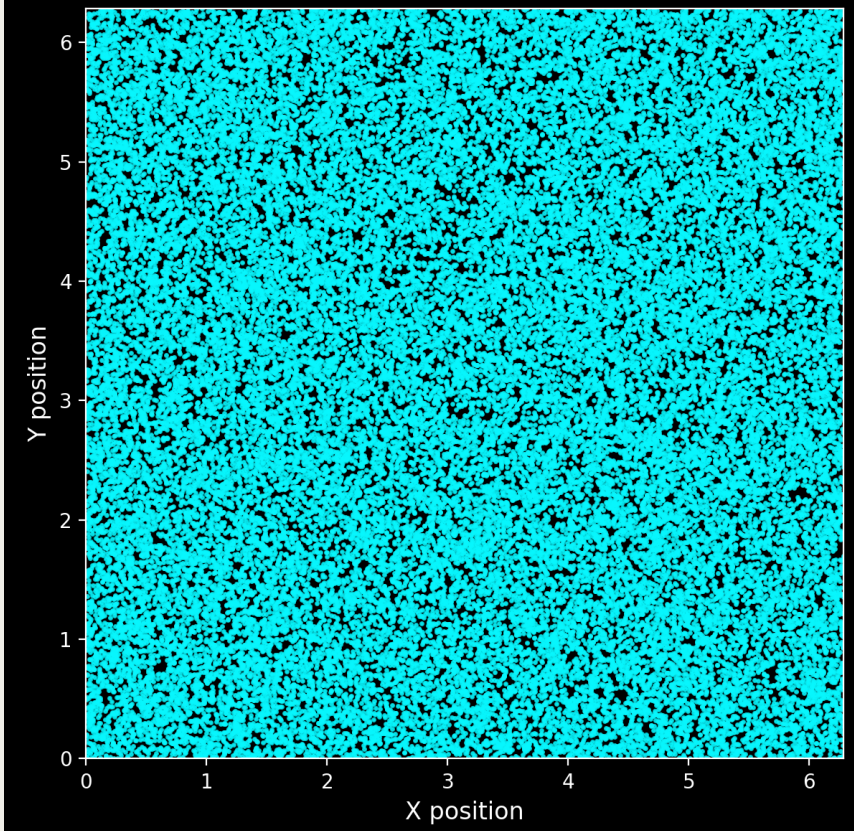
$$\dot{\theta} = \sin(x)\sin(y)$$
$$\dot{x} = \sin(x)\cos(y) + v_s\cos(\theta)$$
$$\dot{y} = -\cos(x)\sin(y) + v_s\sin(\theta)$$

- Studied the combined effect of flow advection and self-propulsion on particle trajectories.
- Orientation evolved through Jeffrey's equation.
- Classified motion using Mean Squared Displacement (MSD).
- Detected chaos through Lyapunov Exponents.



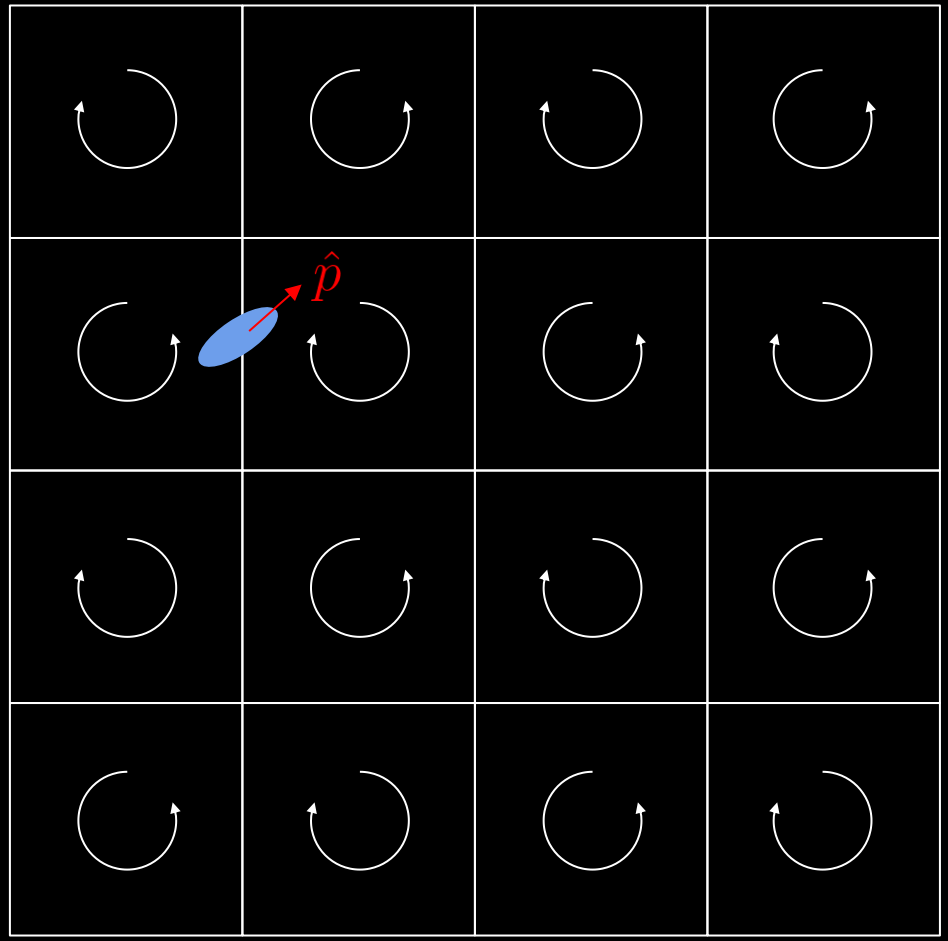


"Active particles show diverse motion regimes and reveal chaotic behavior in structured flows."

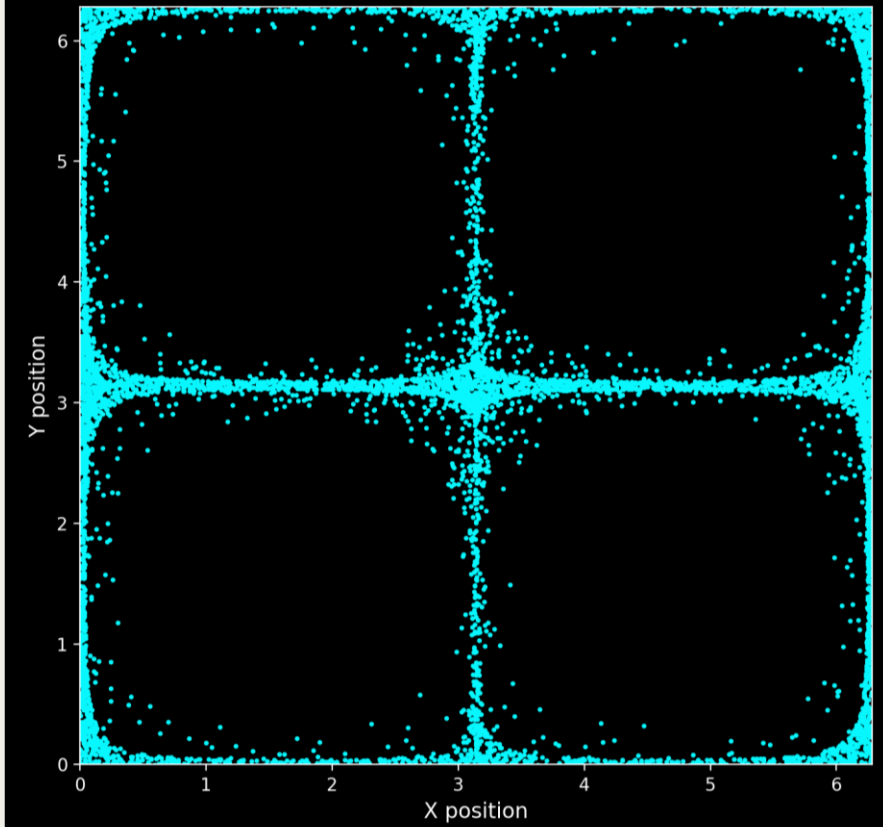


When Shape = Sphere (α = 0):

- All particles are moving in a regular flow. Since the shape is a perfect sphere, it doesn't interact with the flow in a special way.
- As a result, particles stay evenly spread out across the entire space.
- There's no clustering or gathering — the distribution of particles is **uniform**.
- This matches what we expect from Liouville's theorem: the system preserves volume in phase space.

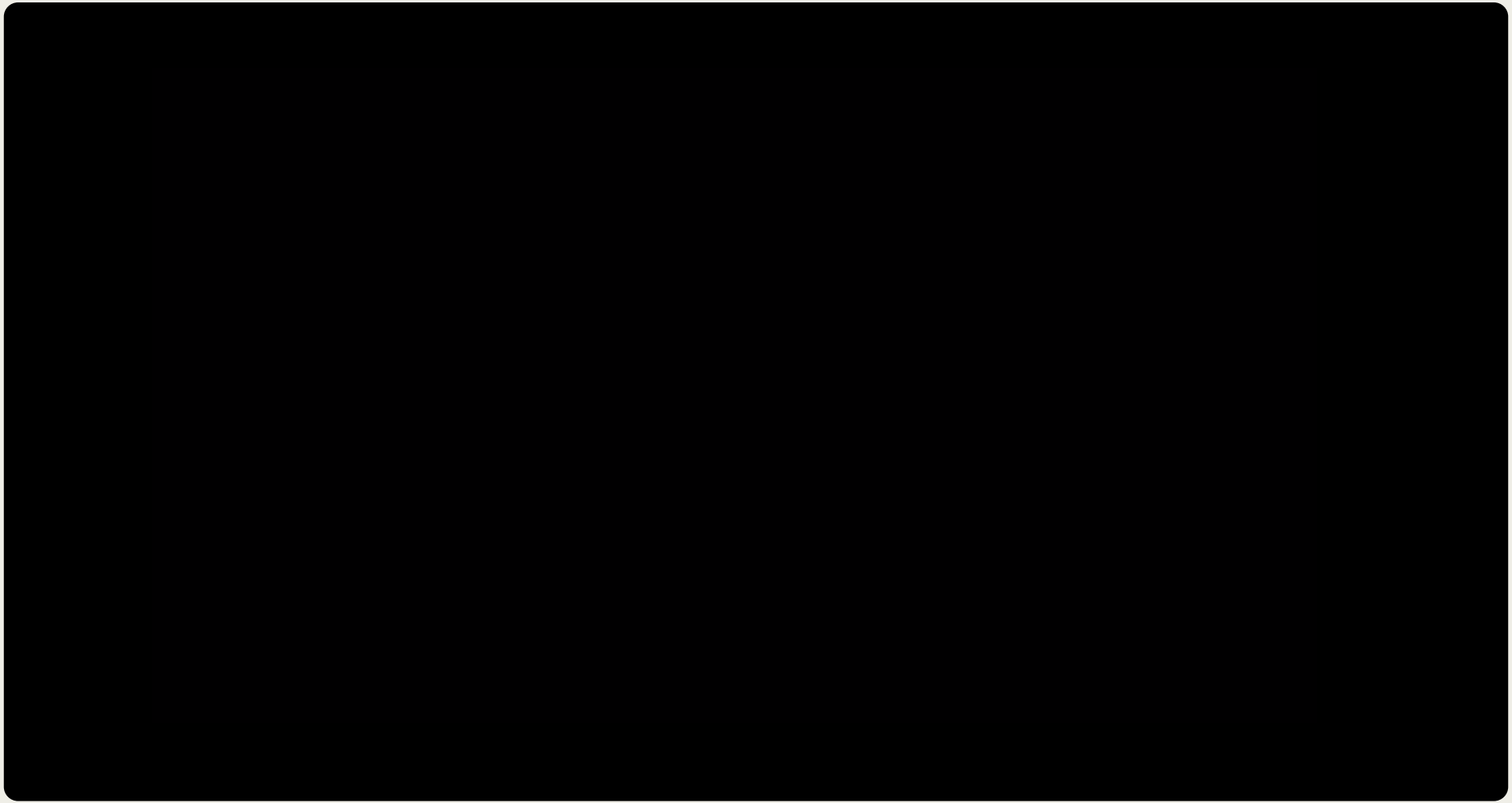


The alignment parameter (eccentricity) α characterizes the shape of the ellipse . For a sphere , γ = 1



When Shape is Ellipsoidal ($\alpha = 0.99$), $v_s = 0.095$

- The particles are stretched and feel the effect of the flow more strongly.
- This shape makes them turn and align with certain flow directions.
- As a result, many particles get pulled toward the edges of the vortex cells.
- You can see **clear clustering** near the boundaries, showing how shape affects motion.



Why Time Dependence Matters?

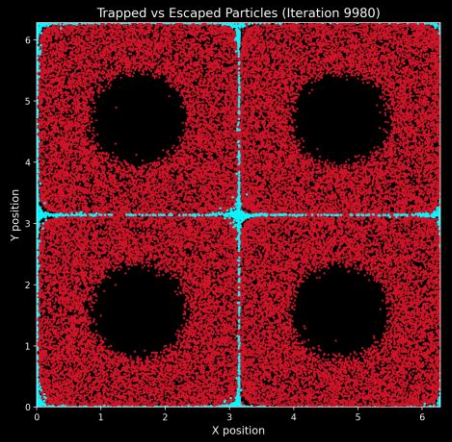
In steady flow, particles get stuck inside vortex cells. When the flow oscillates in time, the borders between cells move.

What Happens?

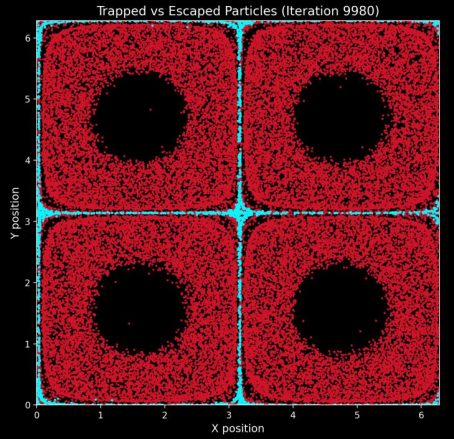
These moving boundaries make it easier for particles to cross into other cells. Even particles with low speed can escape if the flow is time-dependent.

Key Effect:

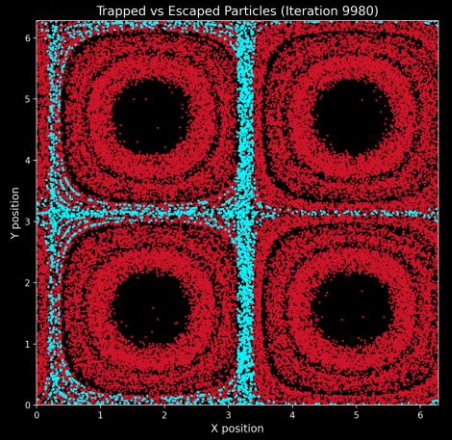
The more the flow oscillates (higher B), the more the trapping barriers are weakened. Particles spread over a larger area and mix better.



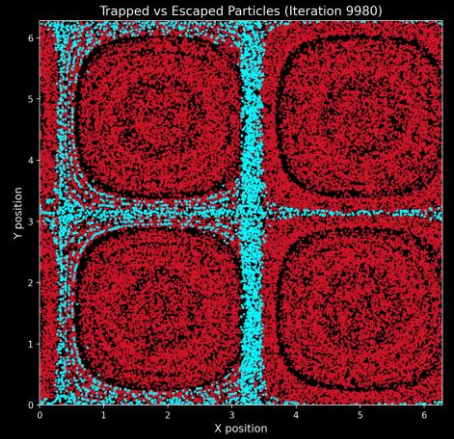
B = 0.15



B = 0.5



B = 3.5



B = 5.0

Rotational Diffusion

What is Rotational Diffusion?

- Particles in fluid randomly change direction over time.
- This happens because of collisions with water molecules.

How It's Included in the Equation?

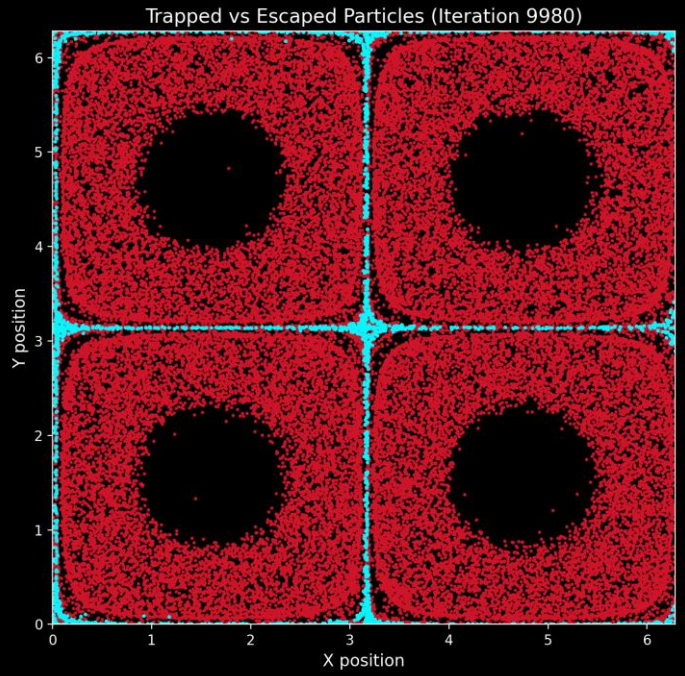
Noise is added directly to the orientation (theta) equation:

$$\dot{\theta} = -\alpha \cos x \cos y \sin(2\theta) + \sin x \sin y + \sqrt{2D_r}\,\eta(t)]$$

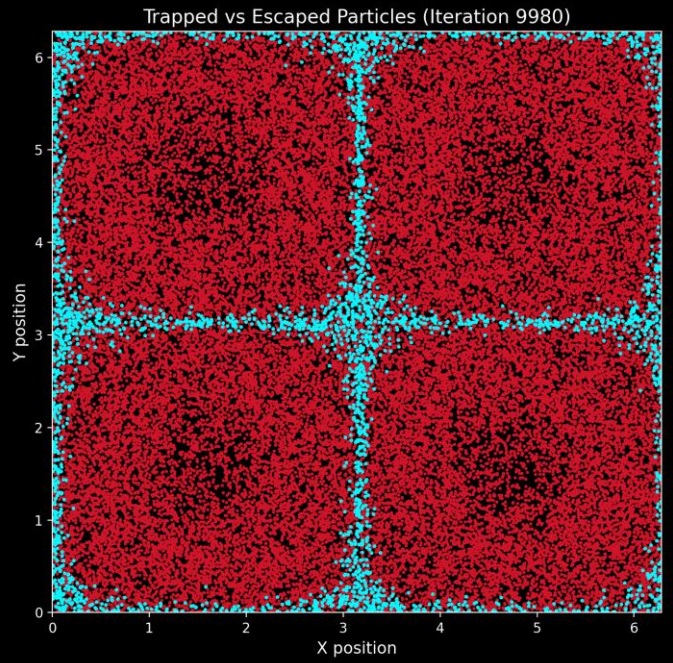
$\eta(t)$:Gaussian white noise(mean=0,variance=1)

D_r :Rotational diffusion constant(controls noise strength)]





B = 0.5 , D_r = 0.0



B = 0.5 , D_r = 0.1

Chirality

What is Chirality?

- Chiral particles naturally **turn left or right** as they move.
- This happens due to **asymmetric shape or propulsion**.

Why It Matters:

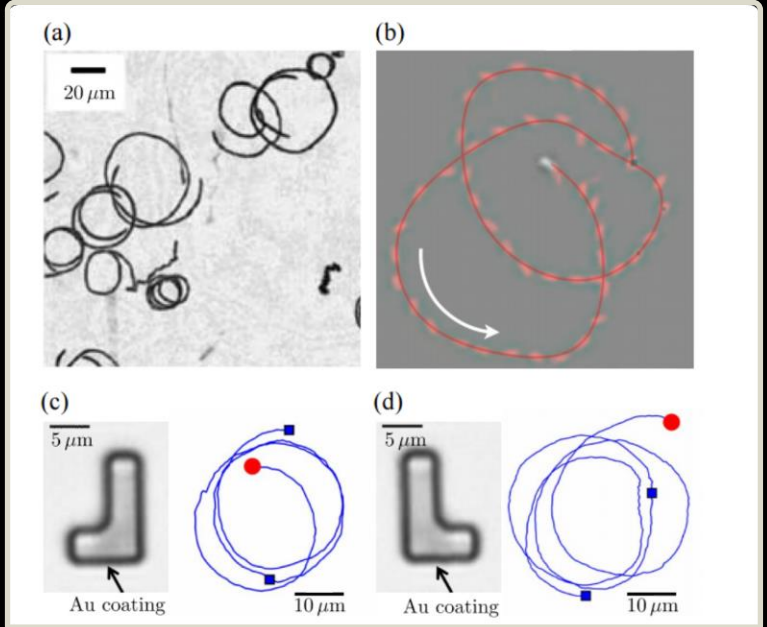
- Chirality breaks the symmetry of motion.
- Particles don't follow straight paths — they trace out **spirals or circles**.
- This affects how they interact with flow structures.

How It Affects Distribution:

•With chirality, particles favor certain **rotational directions**.

•This causes **non-uniform sampling** of the particles.

•Opposite chirality can lead to mirrored trajectories.



•(a,b) Circular swimming patterns of E. coli bacteria

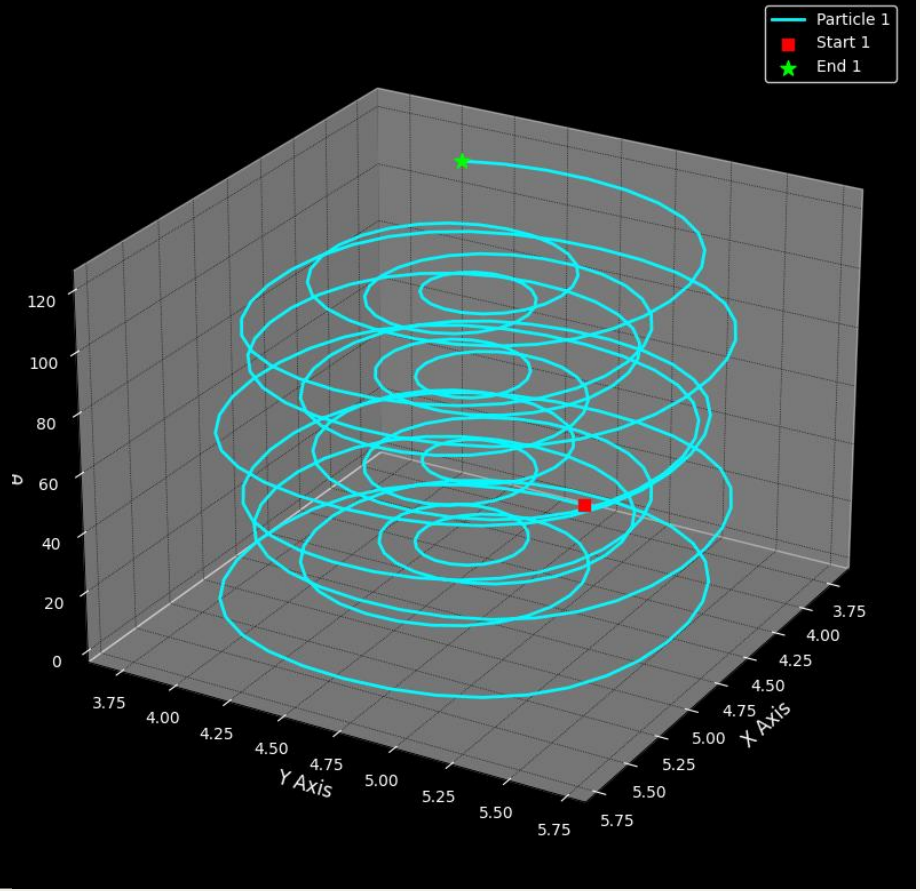
•(c,d) L-shaped particles with different handedness show different path spirals

Chiral Trajectories in Phase Space

What is Phase Space?
•Combines position and orientation into a 3D space.
•Each particle's motion is a **path in this space**.

With Chirality:
•Particles don't just move — their **orientation keeps rotating**.
•This creates **spiral or helical trajectories** in space.

What the Trajectories Show:
•Chiral particles **wind continuously** in orientation.
•The winding direction depends on the **sign of chirality (C)**.

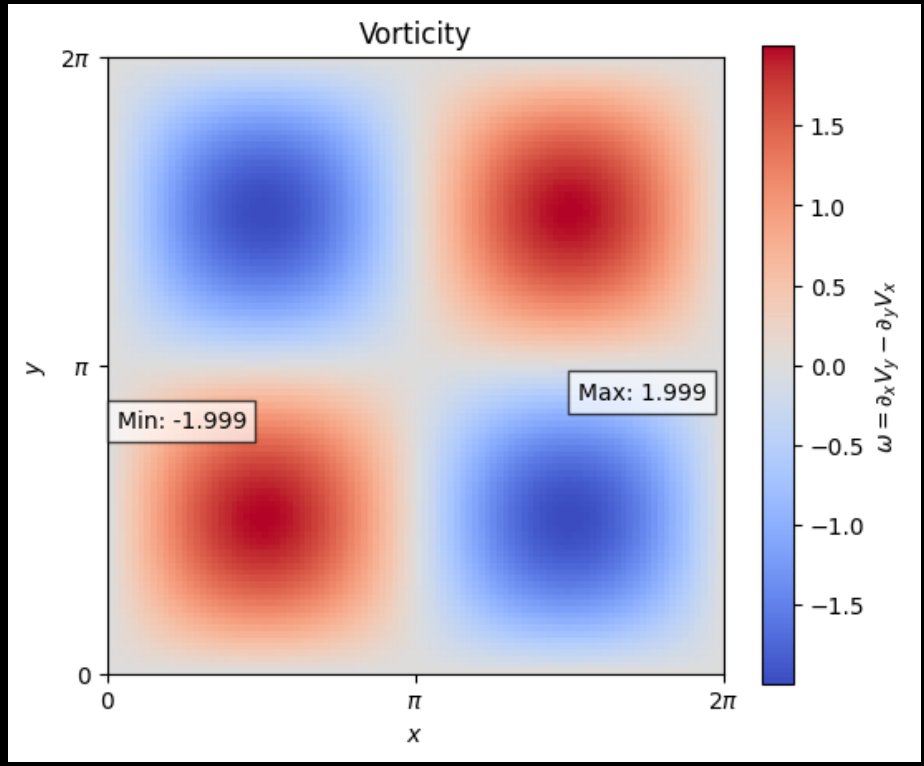


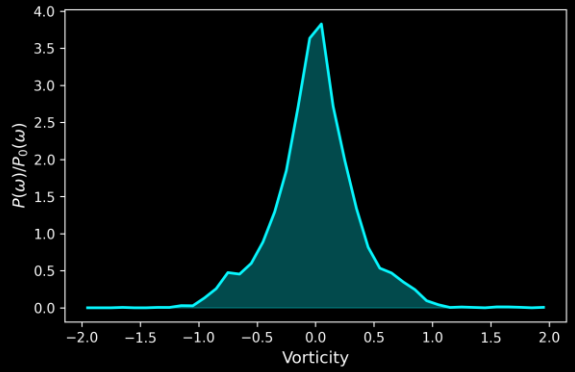
Vorticity

To understand particle behavior, we look at **where they accumulate most** in the flow.
This reveals **long-term motion tendencies**.

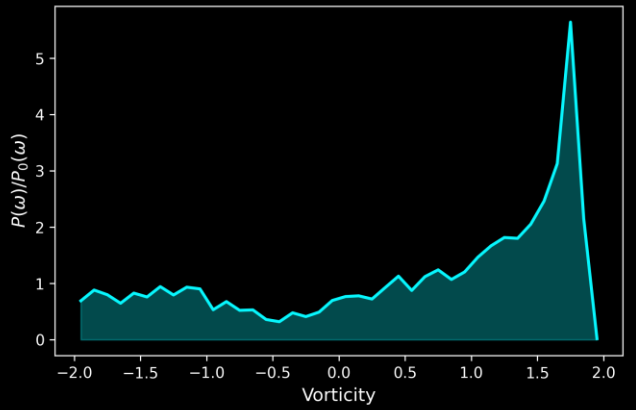
Standard Method:
One way is to compute the **radial distribution function**.
It tells us how likely particles are to be found at different distances.

A Smarter Way — Vorticity Sampling:
Instead of distance, we ask: **what kind of flow are the particles in?**
Vorticity (ω) tells us how much rotation the fluid has at a point.
By tracking the **vorticity at each particle's location**, we learn what kind of flow they prefer.

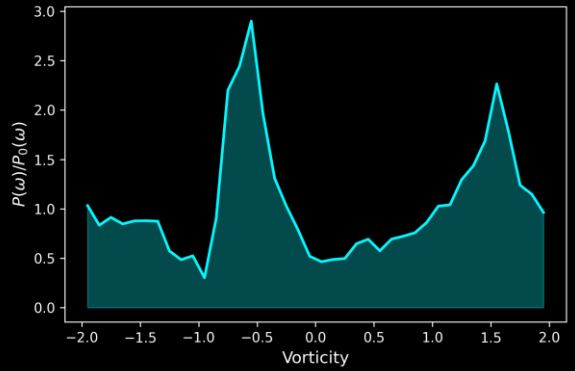




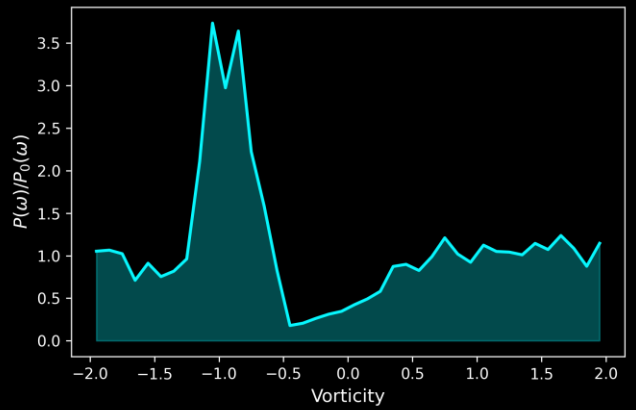
C = 0.0



C = 0.22



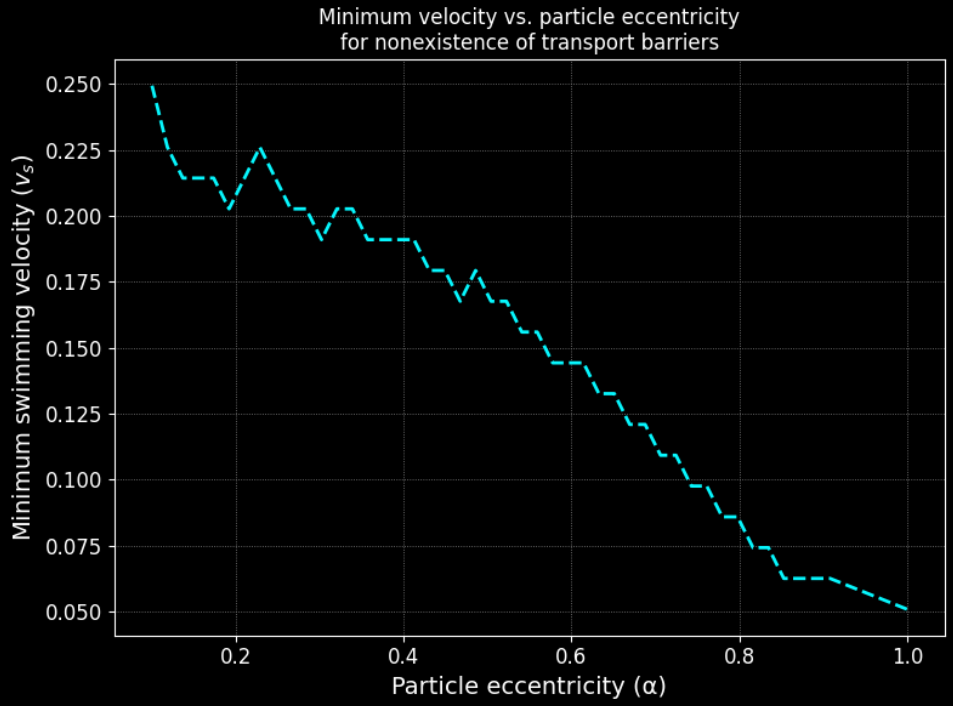
C = 0.67



C = 1.0

Minimum Speed to Escape a Vortex

- To escape a vortex, a particle must swim faster than a certain minimum speed.
- This minimum speed depends on the shape of the particle, given by α.
- In this study, we define escape as the ability of a fraction of particles (say 80% of particles) to leave their initial cell.
- Elongated particles align better with the flow, making it easier for them to pass through flow barriers.
- So, the minimum speed drops as α increases.



"Even a small swimming speed can help elongated particles escape from the vortex ."

Conclusion and Key Insights

1. Shape Drives Transport Efficiency
•Ellipsoidal particles align with flow and escape more easily than spheres, needing less propulsion.

2. Self-Propulsion and Shape Work Together
•Elongated shapes lower the swimming speed needed to overcome trapping barriers.

3. Time-Varying Flows Boost Mixing
•Oscillating flows break cell boundaries, helping even slow particles to move across regions.

4. Noise Enables Uniform Spread
•Rotational diffusion disrupts trapping. When $D_r \geq U_o/L$, particles spread evenly.

5. Chirality Skews Motion and Vorticity Use
•Chiral swimmers favor certain vortices, breaking symmetry and aiding directional sorting.

"Ellipsoidal particles excel at navigating flows — their shape enables smart, efficient transport."

Thanks!

Do you have any questions?